



Dielectric properties and surface potential decay characteristics of low density polyethylene/ZnO microvaristor composites

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Abstract

This paper focuses on the effects of submicron ZnO microvaristors on the microstructure and electrical properties of low density polyethylene (LDPE) based composites. The submicron ZnO microvaristors are obtained by ball milling method and are used as a filler of LDPE based composites. The submicron composites with the filler loading of 0, 10, 20, 30 and 40 wt% are prepared by melt-blending method. The morphology, crystallization behavior, dielectric properties and surface potential decay characteristics are characterized by scanning electron microscope, differential scanning calorimetry, dielectric spectrum and surface potential decay measurement, respectively. Experimental results show that the average speed of the surface potential decay characteristics of LDPE based composites increases distinctly with the increasing filler loading. The submicron ZnO microvaristors are in favorable to suppress the charge accumulation of LDPE based composites and can be used to regulate the charge distribution in electrical and electronic equipment.

1 Introduction

Charge accumulation in the bulk or on the surface of polymeric insulators may distort the electric field distribution, cause the partial discharge, and accelerate the aging of polymeric insulators [1, 2]. Charge accumulation can be suppressed by the regulation of trap distribution in the polymeric insulators [3–5] or by the improvement of the conductivity of the polymeric insulators [6–8]. For example, nano-particles (such as, MgO [3] and ZnO [4, 5]) can introduce a large amount of deep traps into low density polyethylene (LDPE) nano-composites and cause the decrease in charge mobility. As a result, the charge formation and accumulation is suppressed. In contrast, the semiconducting micro-particles (such as micro-sized SiC [6, 7] and ZnO microvaristor [8]) and nano-particles (like nano-sized SiC used in our previous work [9]) can improve the nonlinear conductivity of polymeric insulators. This kind of polymeric insulators shows an excellent electrical insulating property under a very low electric field, while its nonlinear conduction characteristic causes the increase in conductivity under

a high electric field to dissipate the charges accumulated in polymeric insulators.

ZnO is an important inorganic semiconducting material and is usually synthesized by solid stage reaction [10], microwave assisted ultra-sonicated precipitation [11] and sol–gel technique [12], etc. ZnO microvaristors are made from pure ZnO particles by sintering with other metal oxide (such as Bi₂O₃, Sb₂O₃ and MnO₂) at high temperature. ZnO microvaristors are usually used as the filler of nonlinear field grading materials [8], which shows a strong nonlinear conduction characteristic. This kind of nonlinear field grading materials is potentially applied to control the electric field distribution in electrical and electronic applications, such as cable ends, stator bar end of high voltage generator, bushings and power electronic modules. Nowadays, ZnO microvaristors are usually used in the form of large-size particles with an average of 50–100 μm [13]. In order to reach a good nonlinear conduction characteristic, the filler loading of the micro ZnO microvaristors usually exceed 40 vol% (about 75 wt%) [13]. Such high filler loading is unfavorable to material processing and results in the degradation of mechanical properties. It has been reported that the introduction of submicro- and nano- particles into polymeric matrix can improve the electrical and morphological properties [14, 15]. For example, Tjong et al. have proved that the resistivity of LDPE/ZnO nano-composites has a much lower percolation threshold than that of LDPE/ZnO micro-composites

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[14]. Similar observations of the percolation threshold for LDPE/ZnO nano- and submicro- composites have been found by Hong et al. [15]. Decrease of filler content in LDPE based nano- and submicro- composites can improve process ability of these materials [16]. Thus, the submicron ZnO microvaristors are applied as filler to improve the material processing of LDPE based composites. In addition, as ZnO nano-particles and microvaristors can suppress the charge accumulation in polymeric composites [4, 5, 8, 13], the submicron ZnO microvaristors may also be beneficial to suppress the charge accumulation of LDPE based composites and can be used to regulate the charge distribution in the insulating materials in electrical and electronic equipment.

LDPE is a typical insulating material with a low dielectric loss factor, lower relative permittivity and higher resistivity, and it is widely used in high voltage equipment due to the excellent electrical and mechanical characteristics, but it is usually threatened by the charge accumulation [3–5], which can occur in the insulating materials under high electric field in any electrical and electronic equipment. In this paper, LDPE was selected as the polymeric matrix and was modified by submicron ZnO microvaristors to suppress the charge accumulation. The submicron ZnO microvaristors with an average diameter of about 2 μm were used as fillers. LDPE/ZnO microvaristor composites were prepared and their dielectric properties as well as surface charge decay characteristics are discussed. The experimental results show that the average speed of surface potential characteristics increases with the increase in filler loading. The composites are beneficial to the surface charge dissipation.

2 Experimental details

2.1 Materials and preparation

An additive-free LDPE (LD200 with a density of 0.922 g/m³) was purchased from China Petroleum & Chemical Corporation and was used as the polymer matrix. ZnO microvaristor particles were commercially available with a purity degree of 98% and had an original average diameter of about 80 μm . The purchased micro ZnO microvaristor particles were smashed in a QM-3SP04 type planetary ball mill at a revolution rate of 200 rpm for 12 h under room temperature. Thereafter, the submicron particles were obtained by means of filtering, washing and drying the smashed particles. The size distribution of the submicron ZnO microvaristor particles was shown in Fig. 1. Accordingly, we can learn that the submicron ZnO microvaristor particles have an average diameter of 2–2.3 μm . The submicron ZnO microvaristor powders were used as the filler.

The polymer matrix was melt-blended with the filler via an open mill for 30 min. The composites were then put

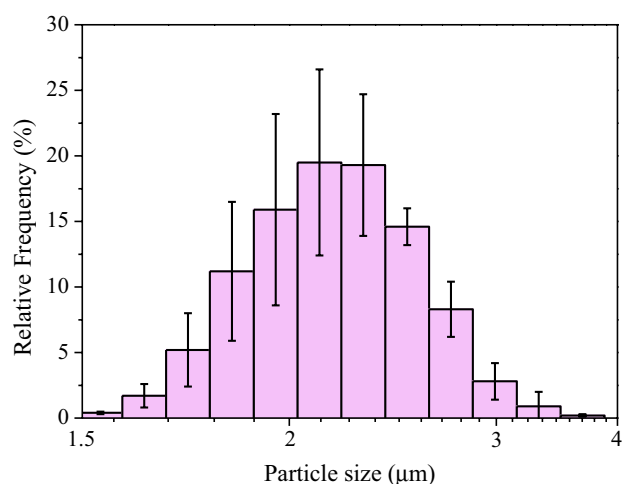


Fig. 1 Size distribution of ZnO microvaristor particles

into mould and were pressed through a plate vulcanizing machine at 130 $^{\circ}\text{C}$ for 20 min. After natural cooling, the specimens of LDPE/ZnO microvaristor composites with a thickness of about 0.2 mm were obtained. The filler loading were 10, 20, 30 and 40 wt%. In addition, pure LDPE specimens were also prepared for comparison.

2.2 Characterizations

A ZS model of Malvern Zetasizer Nano apparatus was applied to characterize the size distribution of the submicron ZnO microvaristor particles. Scanning electron microscope (SEM) of KEYENCE VE-9800 was used to observe the morphology of LDPE/ZnO microvaristor composites. Differential scanning calorimetry (DSC) of Mettler DSC822e was used to measure the crystallization behavior of LDPE/ZnO microvaristor composites. The DSC measurements were carried out from 30 to 150 $^{\circ}\text{C}$ at a rate of 10 $^{\circ}\text{C}/\text{min}$. A Novocontrol Concept 80 equipment was used to characterize the dielectric properties of the specimens at 25 $^{\circ}\text{C}$ in the frequency range of 0.1–10⁶ Hz. A surface potential decay measurement system as described in [17] consists of a needle electrode, a mesh grid, a grounded electrode and an electrostatic voltmeter of Trek P0865, and was used to determine the surface charge decay characteristics of LDPE/ZnO microvaristor composites. The needle electrode and the mesh grid were supplied by DW-SA303-3ACF1 and DW-SA153-3ACF1 type high-voltage dc power supplies made in China, respectively. In this work, a group of dc high voltage of –4 and –14 kV was applied to the mesh grid and the needle electrode, respectively. The charging time was 3 min. All of the surface charge decay measurements were carried out at 25 $^{\circ}\text{C}$ with a relative humidity of about 45%.

3 Results and discussions

3.1 Morphology and crystallization behavior

The SEM images of pure LDPE and 30 wt% LDPE/ZnO microvaristor composites are shown in Fig. 2, respectively. Compared with the uneven section in pure LDPE as shown in Fig. 2a, b shows that the submicron ZnO microvaristors are almost uniformly distributed in LDPE matrix.

Figure 3a shows the DSC thermograms of LDPE/ZnO microvaristor composites with different filler loadings, respectively. When the filler loading increases from 0 to 40 wt%, the melting peak of the DSC thermograms as shown in Fig. 3a shifts slightly around 110 °C. By contrast, the amplitude and width of melting peak alters greatly. It can be seen that, for 40 wt% LDPE/ZnO microvaristor composite, the amplitude of its melting peak decreases almost by half and the width of the melting peak narrows down in comparison with that of pure LDPE. This indicates that

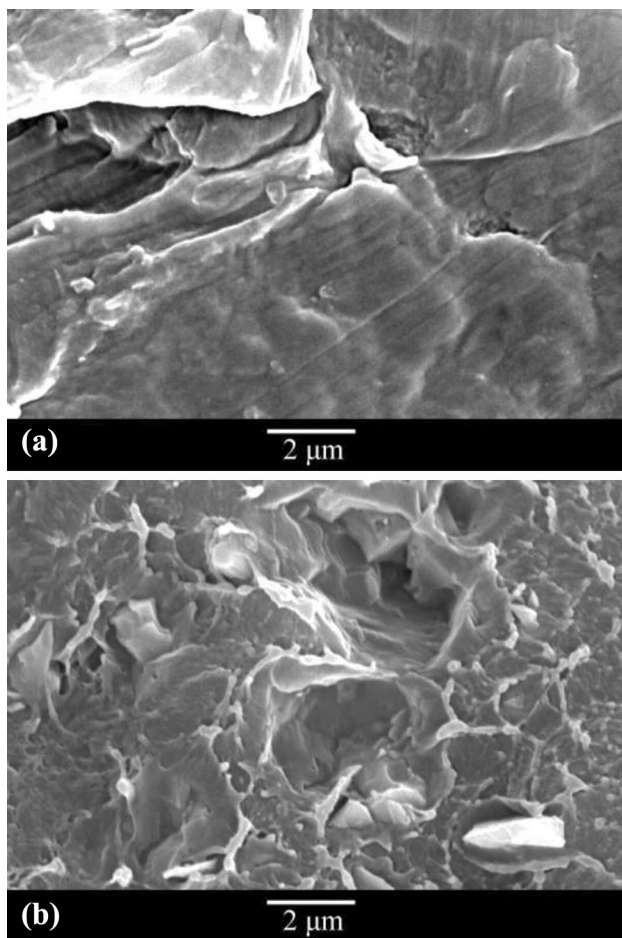


Fig. 2 SEM images of LDPE (a), and 30 wt% LDPE/ZnO microvaristor composite (b)

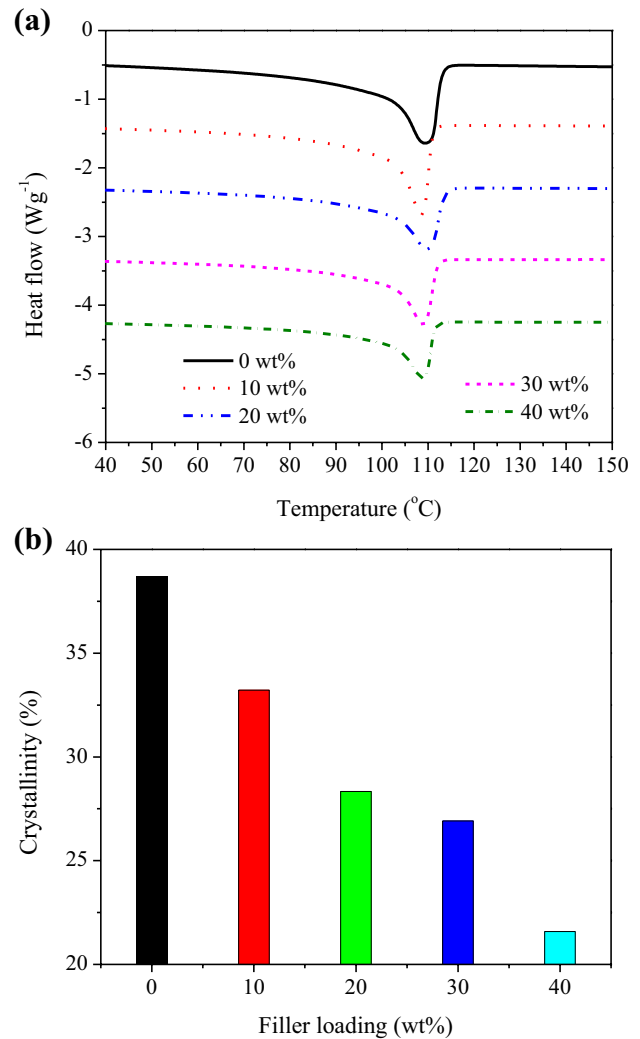


Fig. 3 DSC thermograms of LDPE/ZnO microvaristor composites (a), and crystallinity of LDPE/ZnO microvaristor composites (b)

the filling ZnO microvaristors can affect the crystallization behavior of LDPE composites.

According to the total enthalpy method [18], which is usually used to characterize the crystallization behavior, the crystallinity (X_c) of polymeric insulators can be defined as below.

$$X_c = \frac{\Delta H_m}{\Delta H_0} \times 100\% \quad (1)$$

where ΔH_m is the melting enthalpy in the melting process of polymeric insulators and can be obtained from the DSC thermogram. ΔH_0 is the standard melting enthalpy when the crystallinity of the polymeric insulators is 100%. For LDPE, the typical value of ΔH_0 is 287.3 J/g [18]. Figure 3b shows the crystallinity of LDPE/ZnO microvaristor composites. It can be learnt that the crystallinity of pure LDPE is

38.7%, while as the filler loading increases, the crystallinity of LDPE/ZnO microvaristor composite decreases gradually. When the filler loading is 40 wt%, the crystallinity reduces nearly by 44% compared with pure LDPE. As LDPE is a typical semi-crystalline polymer and usually consists of a great number of crystalline and amorphous regions [19, 20], the decreasing crystallinity in LDPE based composites suggests that the number of crystalline regions reduces, whereas that of amorphous regions increases.

The effects of inorganic fillers on the crystallization behavior of polymers have been studied for many years. Some of these studies have shown that the presence of filler particles can improve the crystallization behavior, resulting in the increase of crystallinity [21]. Other studies demonstrated that the fillers did not change the crystallization behavior of polymers [22]. Besides, the crystallinity of polymers may be reduced because of the hindrance role of filler particles to crystallization behavior [18]. According to the classical Hoffman-Lauritzen model [23], if the filler particles act as a heterogeneous nucleation, the crystallization active energy would decrease and the spherulitic growth rate increases. Meantime, the polymer chains around the filling particles arrange regularly, and the crystallinity of polymeric composite increases. For instance, the crystallinity of LDPE/phenolphthalein composites increases by 11% when the filler loading increase from 0 to 0.4 wt% [18]. However, the crystallinity may reduce because a large number of the filling particles hinder the movement and arrangement of polymer chains, when the filler loading is relatively high. For example, the crystallinity of 5 wt% LDPE/phenolphthalein composite is 34.5% and is much smaller than that of 0.4 wt% LDPE/phenolphthalein composite [18]. Similar variation of the crystallinity against the filler loading has also been observed in the LDPE/alumina nano-composites [24]. Obviously, the ZnO microvaristors has a larger average diameter compared with previous nano-particles. The large particle size of ZnO microvaristor may make the surrounding LDPE molecular chains difficult to move and arrange regularly, so the crystallinity of LDPE/ZnO microvaristor composite reduces even when the filler loading is very low (e.g., 10 wt%), compared with pure LDPE.

3.2 Dielectric properties

The variations of the real part (ϵ') and imaginary part (ϵ'') of complex permittivity of LDPE/ZnO microvaristor composites against the applied frequency (f) are shown in Fig. 4a, b, respectively. In Fig. 4a, for pure LDPE, ϵ' varies little with the increase of the frequency. By contrast, for the 40 wt%, ϵ' decreases greatly with the increasing frequency, especially in the range of 0.1 through 10^3 Hz. In addition, the real part of complex permittivity increases

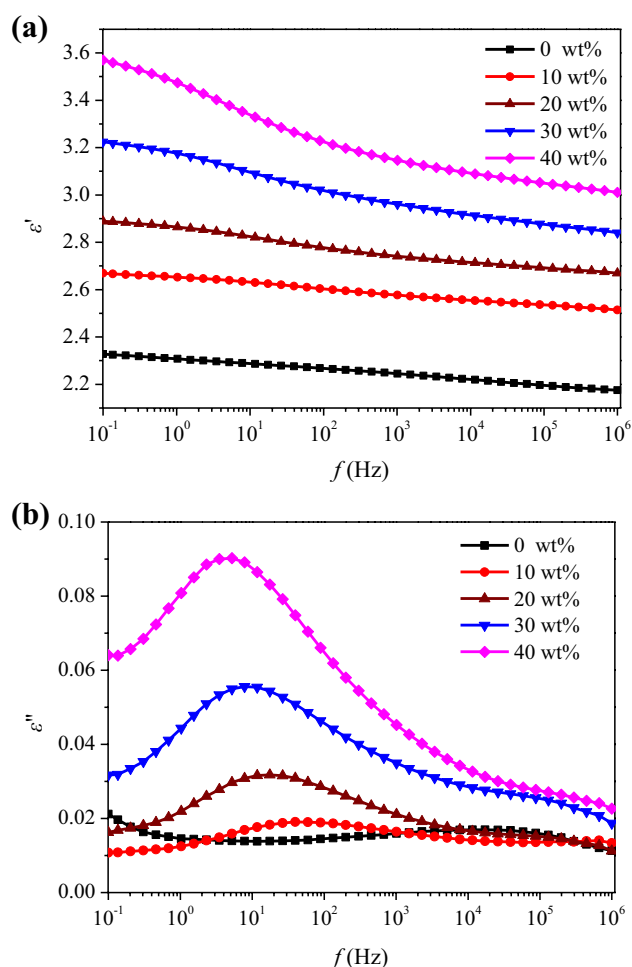


Fig. 4 Relationships between the complex permittivity and frequency of LDPE/ZnO microvaristor composite. Real part of complex permittivity (a), and imaginary part of complex permittivity (b)

with the increase in filler loading at the same frequency. In other words, an extra relaxation occurs in LDPE based composites, compared with pure LDPE. Similar observations have been found in nano-composites with higher filler loading [25, 26]. Thus, it is reasonable to believe that the extra relaxation process is caused by the interface between the fillers and LDPE matrix.

In Fig. 4b, LDPE/ZnO microvaristor composites with 10, 20, 30 and 40 wt% have a distinct loss peak in low frequency range (i.e., 0.1– 10^3 Hz), compared to pure LDPE. The formation of this loss peak in each ϵ'' - f curve for LDPE based composites is closely related to the extra relaxation polarization caused the interface between the fillers and LDPE matrix. In addition, the amplitude of the distinct loss peak as shown in Fig. 4b increases with the filler loading, which indicates that the number or volume fraction of the interface increases with the increasing filler loading.

3.3 Surface potential decay characteristics

Figure 5a shows the surface potential decay characteristics of negatively charged LDPE/ZnO microvaristor composites, and the surface potential decreases with time for each LDPE composite. The decay degree of the surface potential decay characteristic can be evaluated by an average speed (v_{as}), defined as below [17].

$$v_{as} = \frac{(U_0 - U_t)}{tU_0} \times 100\% \quad (2)$$

where U_0 is the initial surface potential, and U_t is the surface potential at time t . The variation of the average speed with respect to the filler loading is shown in Fig. 4b. It can be seen that the average speed increases with the filler loading. The average speed increases by 57% when the filler loading is up to 40 wt%, compared with pure LDPE. Since the

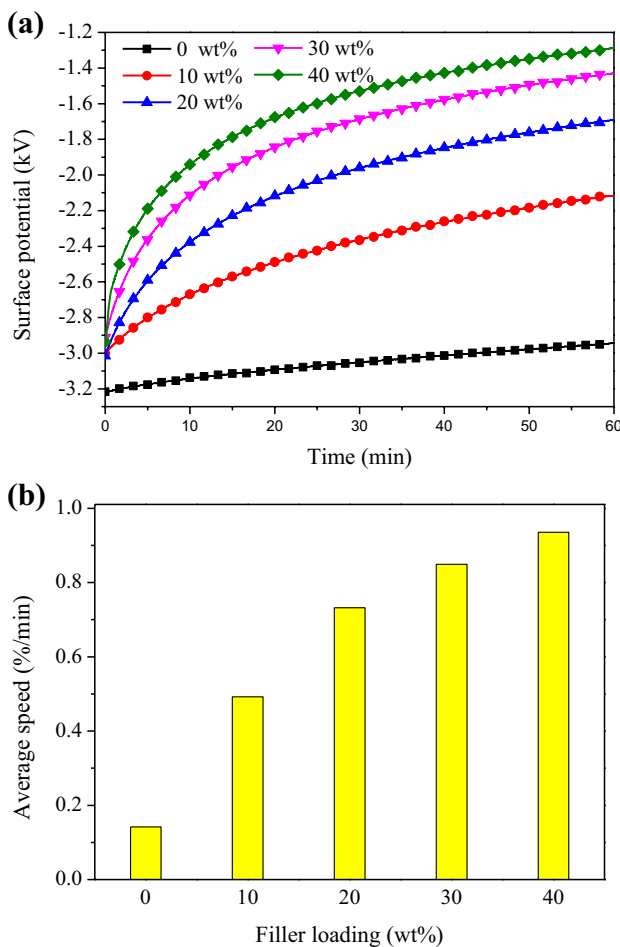


Fig. 5 Surface potential decay characteristics of LDPE composites (a), and the relationship between the average speed and the filler loading (b)

surface charge density is proportional to the surface potential [19], Fig. 5b also suggests that the surface charges accumulated on the negatively charged LDPE/ZnO microvaristor composites also decays much faster with the increase in filler loading.

Trap distribution of polymeric insulators can be obtained from the surface potential decay characteristic when the charge carriers' transit time is shorter than their mean detrapping time [20], and can be defined as follows [20].

$$E_t = kT \ln(vt) \quad (3)$$

$$N_e(E_t) = \frac{n_{e10}r_{de1}t \exp(-r_{de1}t) + n_{e20}r_{de2}t \exp(-r_{de2}t)}{f_0(E_t)} \quad (4)$$

where E_t is the energy level of trap distribution. $N_e(E_t)$ is the trap density at the energy level of E_t . n_{e10} and r_{de1} is the initial density and the detrapping probability of trapped charges in shallow traps, respectively. n_{e20} and r_{de2} is the initial density and the detrapping probability of trapped charges in deep traps, respectively. n_{e10} , n_{e20} , r_{de1} and r_{de2} can be obtained from the surface potential characteristics. k is the Boltzmann's constant. T is the temperature. v is the attempt to escape frequency. $f_0(E)$ is the initial occupancy of the traps.

According to Eq. (3) and (4), the trap distributions of LDPE/ZnO microvaristor composites with 0, 10, 20, 30 and 40 wt% are obtained as shown in Fig. 6. Two distinct peaks (i.e., Peak A and Peak B) can be clearly observed in each trap distribution. Peak A has a lower energy level than Peak B. Consequently, Peak A represents the shallow traps, while Peak B represents the deep traps. The energy levels of shallow and deep traps decrease when the filler loading increases. In addition, the trap density of shallow traps also

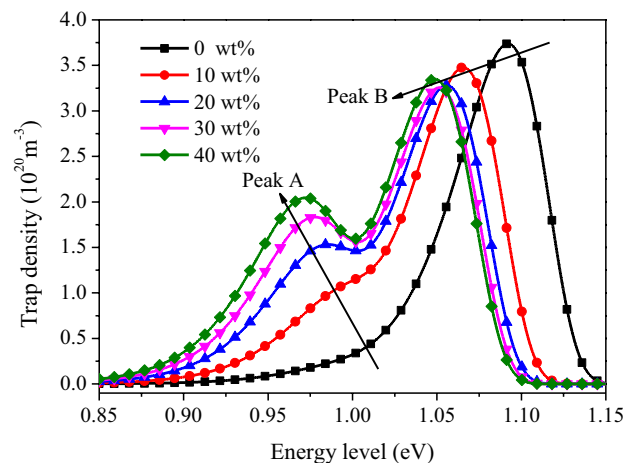


Fig. 6 Trap distribution of LDPE/ZnO microvaristor composites with different filler loading

increases with the filler loading, while that of deep traps nearly reduces with the increase in filler loading. Similar phenomena have been reported for LDPE/ZnO nanocomposites and micro-composite in our previous work [20].

The multi-core model is usually used to represent the interface between the nanoparticle and polymer matrix in nanocomposites, and it consists of a bonded layer, a bound layer and a loose layer [27]. In the bonded and bound layers, the deeper traps are usually formed due to the ordered molecular structure. By contrast, in the loose layer, the shallower traps are formed due to the amorphous structure in the loose layer. The interface between the submicron particle and polymer matrix can be also described by the multi-core model. But there is only a much weaker interfacial interaction in submicron composites than that in nanocomposites, where the interfacial interaction is enhanced by the coupling agent. According to the multi-core model, there are a few deeper and shallower traps in the interface between the particle and polymer matrix, so there are two episodic peaks of trap distribution in the LDPE/ZnO microvaristor composites. One (i.e., Peak A) represents the shallow traps while the other (i.e., Peak B) represents the deep traps. For pure LDPE, in its crystalline regions, the deep traps may be formed due to some chemical defects [19, 20]. While in the amorphous regions, the shallow traps may be generated by some physical defects. Consequently, two episodic peaks exist in the trap distribution of pure LDPE.

In submicron composites, the shallower traps in the loose layer may be added to the original traps or replace the original traps in polymer matrix, leading to the shift of the original traps to much shallower energy level. With the increase in filler loading, the dielectric properties suggest that the volume fraction of the interface between the submicron particle and LDPE matrix increases, resulting in the increasing volume fraction of the loose layer. Moreover, DSC results suggest that the number of crystalline regions reduces, whereas that of amorphous regions increases, when the filler loading increases. As the shallow traps usually exist in the loose layer around the submicron particles and in the amorphous regions of the polymer matrix, the LDPE/ZnO microvaristor composites with higher filler loading have a much lower energy level of shallow and deep traps than those with lower filler loading. In addition, the increasing volume fraction of the loose layer can also cause the increase of shallow trap density and the decrease of deep trap density.

As is known, the charges are more likely to be motivated from shallow traps than from deep traps. As a result, the charges captured by traps in the LDPE/ZnO microvaristor composite with higher filler loading need less energy to get out of traps, compared to the one with lower filler loading. In other words, for the surface charge decay characteristics, the LDPE based composite with higher filler loading has a large average speed than the one with lower filler loading.

4 Conclusions

The submicron ZnO microvaristors play an important role in the microstructure and electrical properties of LDPE based composites. With the increasing filler loading, the crystallization of LDPE based composites becomes difficult. Compared with pure LDPE, the crystallinity of 40 wt% LDPE based composites reduces almost by 44%. The real part of complex permittivity also increases when the filler loading increases from 0 to 40 wt%, and a distinct loss peak in low frequency range of $0.1\text{--}10^3$ Hz is formed due to the relaxation polarization caused by the interface between the fillers and LDPE matrix. Surface potential measurements show that the composites with higher filler loading have a much larger average speed than those with lower filler loading. For example, the 40 wt% composites have the maximum average speed with a value of $0.94\%/ \text{min}$ among all the LDPE based composites in this work. This suggests that the charge accumulation of LDPE can be improved by the submicron ZnO microvaristors, which are of great potential application in electrical and electronic equipment.

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